

Software Market Dynamics

***Software is changing slower than hardware
advances but faster than the ability of
the typical company to assimilate
change***

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Agenda

- **Operating systems**
- **Database management systems**
- **Client-server application development**
- **Client-server application solutions**
- **Enterprise Systems Management**
- **Opportunities**

Reference Platform

Operating Systems

- Old, proprietary OS's (MVS, VMS) don't matter
- Windows 95 will gain rapid acceptance by 1996 after a rocky start
- Microsoft NT Advanced Server sweeps up from the workgroup towards the enterprise level.
 - When will NT server revenues pass Unix server revenues?
 - Microsoft's proprietary OS ala AS/400
- IBM OS/2 Warp missed the window (pun intended). Can never recover.
- UNIX variants prove able to do single enterprise-size applications
 - Mature market
 - ISVs eye NT as next step

Database Management Systems

- **Relational (RDBMS) is hot-hot-hot**
 - **Estimate 9 RDBMS licenses for every 10 multiuser Unix servers sold**
- **Pure object-oriented (ODBMS) window has closed -- a niche**
- **Existing players get richer**
 - **Object-relational marriage**
 - **Personal RDBMS**
 - **Distributed-administration simplicity**
 - **Gateways to data sources vs. ODBC**
 - **Replication**
 - **Data warehousing is reaching mainstream**

Client-Server Application

Development

- **Client-server will evolve to a distributed computing model by year 2000**
- **Market for development tools is still fragmented**
- **Next big market move is into object-oriented development. Products are already in place, but need infrastructure such as object request brokers.**
 - **Will be very hard for IS to learn**
 - **PC development is not enterprise development!**
- **Common repository for metadata is a missing link**

Client-Server Application Solutions

- **Barriers to entry for Financials, HR, high-end manufacturing (i.e., SAP) are huge**
- **1995 sees mainstream availability of a wide variety of client-server vertical apps**
- **Mainframe/proprietary mini apps are dead**
- **Early C-S ISVs delivered on Unix servers ...**
- **Now, Windows NT is a common option**
- **Microsoft Windows is the client platform-of-choice**
- **But emerging applications create new market opportunities**
 - **Customer Interaction Software**
 - **Mapping**

Enterprise Information Systems Management

- **Distributed systems are rapidly causing administrative chaos. Huge interest in alleviating the IS pain.**
- **EISM is Aberdeen's model for consolidation of key technology areas:**
 - **Network management**
 - **Systems management**
 - **Operations management**
 - **Database management**
- **Many niche players but large suppliers are driving consolidation**

Opportunities in Software

- **Software integration services compensate for IS inability to choose/implement**
- **Object-transitional aids (e.g., Object COBOL, legacy code-wrappers, ORBs, design/code/test tools)**
- **Client-server app capacity planning, performance monitoring, and testing**
- **Making NT servers robust (e.g., OLTP, clusters, Novell migration)**
- **Distribution of applications to second-tier companies (e.g., SAP R3 to \$100m manufacturers)**
- **Document management**
- **Management-by-exception (e.g., artificial intelligence hooked to data warehouse) query tools**